

We thank the Editor and the Reviewers for their constructive comments and suggestions, which we have addressed below. Modifications are indicated in the “Manuscript-tracked.docx” and “Supplementary Materials-tracked.docx” files using track changes. In our answers below, we refer to the corresponding page and line numbers in the “All Markup – showing all revisions inline” display of the tracked files. The “Manuscript-clean.docx” and “Supplementary Materials-clean.pdf” files are the clean versions of the revised manuscript and supplementary materials, respectively.

For clarity, the Reviewers’ comments are presented in black, and our point-by-point responses are provided in blue.

Reviewer #1

This article is a population pharmacokinetics (popPK) study of fluconazole aimed at improving target attainment in critically ill patients. Overall, the research design, methodology, and presentation of results are quite comprehensive. However, there are still some aspects that may need further discussion or clarification. The following are some potential issues and areas to consider:

1. *The study should indicate in the abstract section how to adjust the fluconazole dosage for ICU patients on CRRT, and for patients not on CRRT weighing more than 60 kg.*

Thank you for this important comment. We have now rephrased the results of our abstract to indicate how to adjust fluconazole dosage for these patients as “*A weight-based loading dose and flat maintenance doses of 400mg (off-CRRT) and 800mg (CRRT) predicted $\geq 90\%$ PTA across the weight range*”. [Page 4, lines 112–114]

2. *The study integrates data from eight different studies, which increases the sample size but may introduce heterogeneity. Differences in patient selection, sampling times, and analytical methods across these studies could affect the homogeneity of the data and the reliability of the results. For example, the recording of CRRT (continuous renal replacement therapy) types and parameters may not be consistent across studies. This should be discussed in the limitations part of the study.*

We have now discussed this in the limitations part of our study as “*Third, our study integrates data from eight different studies, which increases the sample size but may introduce heterogeneity. Variations in patient selection, sampling times, and analytical methods across these studies could affect data consistency and the reliability of the results. For example, the recording of CRRT types and parameters may not have been consistent across studies*”. [Page 16, lines 418–422]

3. *Covariate Handling: While stepwise covariate modeling was used to select significant covariates, the interactions between these covariates (e.g., between CRRT status and eGFR) may not have been fully considered. Additionally, the handling of missing data (e.g., through multiple imputation) is reasonable but may introduce additional uncertainty.*

Thank you for this important comment. Regarding interactions between covariates, we have added a sentence to the limitations part of our discussion as “*Fourth, we did not explore potential interaction terms between covariates in our model, which may improve model fit*”. [Page 16, lines 423–424]

Regarding multiple imputation, we have added that multiple imputation “*has been shown to better reflect the uncertainty of the parameter estimate for the missing variable compared to single imputation*” to our methods section [Page 8, lines 200–202].

As we cited two new references to address this comment, our reference list has been updated accordingly [Pages 22–26, lines 546–674].

4. *Threshold Setting for Target Attainment (PTA): Although the study set $fAUC_{0-24}/MIC = 100$ as the target, the MICs of different strains may vary significantly in clinical practice, which could affect the practical significance of the target attainment rate. Moreover, for strains with higher MICs (e.g., $MIC = 4$ mg/L), the proposed dosing regimen may not be sufficient to achieve the target.*

Thank you for the comment. We have acknowledged this in the limitations part of our manuscript as “*Fifth, although the study set $fAUC_{0-24}/MIC = 100$ as the target, MIC values can vary significantly across strains in clinical settings, which may limit the practical relevance of the target attainment rate. For strains with higher MICs (e.g., 4 mg/L), the proposed dosing regimen may be insufficient to achieve the desired exposure*”. [Page 16, lines 424–427]

5. *Feasibility of Dose Adjustment: The study proposes an optimized dosing regimen based on patient total body weight and CRRT status, but frequent dose adjustments may be challenging in actual clinical practice, especially in the ICU setting where patients' conditions and physiological parameters can change rapidly.*

We have also acknowledged this in the limitations part of our manuscript as “*Sixth, although our study proposes an optimised dosing regimen based on total body weight and CRRT status, frequent dose adjustments may be difficult to implement in clinical practice, particularly in ICU settings, where patients' conditions and physiological parameters can change rapidly*”. [Pages 16–17, lines 427–430]

6. *The authors stated in the Results section: “The standard loading dose did not result in clinically acceptable PTA of $\geq 90\%$ when total body weight exceeded 65 kg and 30 kg, in patients off and on CRRT, respectively.” However, in the Abstract, they mentioned: “With standard dosing, only CRRT status and total body weight were clinically relevant, as PTA dropped below 90% for all patients on CRRT, and for patients not on CRRT weighing more than 60 kg.” The authors should clarify whether it is 60 kg or 65 kg, and ensure consistency between the two sections.*

We have corrected our Results section from 65 kg to 60 kg. [Page 12, line 310]

Reviewer #2

Dear authors,

1. *Please display our main conclusion already in the abstract!*

Thank you for the very good point, which has also been suggested by Reviewer #1. We have now added to the conclusion part of our abstract to clarify that our proposed dosing regimen is “*stratified by weight and CRRT status*” [Page 4, lines 115–116]. We have also touched upon the main conclusions in the results section as “*with standard dosing, only CRRT status and total body weight were clinically relevant*” [Page 4, line 110], and “*A weight-based loading dose and flat*

maintenance doses of 400mg (off-CRRT) and 800mg (CRRT) predicted $\geq 90\%$ PTA across the weight range” [Page 4, lines 113–114].

To conform to the 250-word limit, we have made some minor changes in Page 4; lines 93–96, line 100, and lines 111–112.

2. *You applied the CDK EPI equation for estimation of the GFR (eGFR). The eGFR, however, is well established in patients with stable renal impairment, but generally unreliable parameter in critically ill because of changes in renal function during severe illness. Please discuss this problem in the limitations section.*

We have acknowledged this in the limitations part of our manuscript as “*Seventh, we applied the CKD-EPI equation to estimate the eGFR, of which the reliability remains questionable in critically ill patients [41]. However, this is a general problem applicable to all eGFR estimates, and CKD-EPI has been shown to be the least worst [41].*” [Page 17, lines 430–433]

As we cited a new reference to address this comment, our reference list has been updated accordingly [Pages 22–26, lines 546–674].

3. *You have focussed on pharmacokinetic data derived from previous studies and, therefore, you have not assessed toxicity or adverse effects in your present study. only, the risk of trough concentrations above 80 mg/l was calculated. Unlike the label, you do not recommend a reduction of the maintenance dose for patients with a GFR < 30 ml/min. Fluconazole is relatively well tolerated, but it might cause hepatotoxicity as well as QT prolongation at ECG, bearing the risk of life-threatening arrhythmias. Therefore, please discuss safety aspects in the light of the relatively high loading doses you recommend!*

Thank you for your valuable comment. We have now added a paragraph discussing the safety aspects of our proposed dosing regimen in the Discussion section of our paper as “*Unlike the label, we did not recommend a reduction of the loading or maintenance doses for patients with renal impairment (i.e., creatinine clearance ≤ 50 mL/min) [5]. While fluconazole is generally well tolerated, it may cause hepatotoxicity and QT prolongation on electrocardiogram, posing a risk of life-threatening arrhythmias [5]. Therefore, future prospective studies should assess the safety profile of our proposed dosing regimen.*” [Page 16, lines 404–408]

4. *In the legend of figure S12 you mention a “vertical dashed green line” that “illustrates the start of fluconazole infusion”. I cannot see this line. Please clarify?*

We have removed that sentence from the legend of Figure S12. We have also done the same for Figures S10 and S11 to avoid confusion. [Supplementary Materials-tracked.docx file]